Advanced Programming Practice · 21CSC203P

Unit 4 · Session 6 · SLO 2 — Solved Worksheet

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Unit – 4 : Session – 6 : SLO – 2 | Parallel Computing — Architecture Matching

I. Match the Following

1. Shared Memory — Common address space: all processors read/write one global memory.
2. Distributed Memory — Message passing: each node has private memory; data moves via explicit send/receive (MPI).
3. Multicore CPU — Parallel execution units: several cores on one chip execute threads simultaneously.

II. Key Notes

• Flynn's taxonomy context: multicore CPUs and shared-memory machines are typically MIMD systems.
• Hybrid systems combine both: clusters of multicore nodes use shared memory inside a node and message passing across nodes.